

NORTH PLATTE, NE, USA

WMO: 725620

Lat: 41.1328N

Lon: 100.7000W

Elev: 2783

StdP: 13.28

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 24023

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.7	1.3	-10.4	3.5	-2.0	-5.4	4.6	3.1	30.9	33.8	26.9	35.7	6.4	310	0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.8	96.2	70.4	93.0	70.7	89.8	70.1	75.1	88.0	73.5	86.6	72.2	85.1	11.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.5	128.9	81.3	69.8	121.5	80.1	68.2	115.0	78.7	40.6	88.2	39.1	86.8	37.7	85.0	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.4	24.0	20.2	-13.7	101.5	4.9	4.0	-17.2	104.4	-20.1	106.7	-22.8	109.0	-26.3	111.9		
			-14.1	78.6	4.8	2.0	-17.6	80.0	-20.4	81.2	-23.1	82.3	-26.6	83.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	26.1	28.7	39.0	47.8	58.0	69.3	75.3	72.9	64.1	49.8	37.0	26.9
	DBStd	19.70	10.95	11.41	10.90	9.40	8.57	7.20	5.70	5.73	8.76	9.19	9.69	10.16
	HDD50	3108	741	596	363	148	24	1	0	0	6	115	397	717
	HDD65	6463	1206	1015	807	519	247	38	4	8	121	476	840	1182
	CDD50	2994	0	1	21	81	274	579	785	708	429	109	7	0
	CDD65	877	0	0	0	2	32	166	325	252	95	5	0	0
	CDH74	10502	0	1	34	123	576	2044	3662	2606	1279	170	7	0
	CDH80	4689	0	0	5	25	184	867	1843	1186	537	42	0	0
Wind	WSAvg	8.9	8.2	9.0	10.1	11.3	10.0	9.1	8.0	7.6	8.5	8.9	8.3	8.0
Precipitation	PrecAvg	20.0	0.4	0.5	1.1	2.2	3.3	3.5	3.0	2.1	1.6	1.3	0.5	0.4
	PrecMax	29.6	1.1	2.0	3.0	5.9	8.0	8.7	7.0	6.3	6.0	4.8	1.9	2.6
	PrecMin	10.1	0.0	0.0	0.0	0.1	0.7	0.4	0.2	0.1	0.1	0.0	0.0	0.0
	PrecStd	4.0	0.3	0.5	0.8	1.3	1.7	1.7	1.6	1.4	1.4	1.1	0.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.6	68.8	78.6	83.9	90.9	97.7	102.1	99.0	95.4	85.5	73.9	61.7
		MCWB	45.0	48.2	54.3	57.8	65.5	69.6	70.5	70.1	67.2	60.4	52.4	45.7
	2%	DB	54.7	60.1	71.4	77.4	85.1	93.0	97.0	94.4	91.0	78.7	66.6	54.2
		MCWB	41.4	44.1	51.2	56.6	64.8	70.5	71.5	70.6	67.0	58.5	49.3	41.6
	5%	DB	48.7	53.5	65.2	72.4	80.4	88.9	93.5	90.7	86.5	73.3	60.6	48.7
		MCWB	38.5	41.7	48.7	54.2	62.7	69.2	71.9	70.7	66.2	56.2	46.4	38.6
	10%	DB	43.0	47.2	59.2	66.6	76.0	84.9	90.1	86.9	81.8	67.8	54.4	42.9
		MCWB	35.4	38.1	45.9	51.8	60.3	67.9	71.7	70.3	64.0	54.0	43.8	35.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.9	49.5	58.0	61.5	69.6	75.5	77.9	76.5	72.7	64.5	54.3	46.9
		MCDB	60.6	64.9	69.6	78.3	83.9	88.3	90.4	88.4	86.5	77.7	67.8	60.0
	2%	WB	42.0	45.1	53.5	58.1	66.7	73.0	75.6	74.5	69.9	61.2	50.7	42.2
		MCDB	54.0	58.1	68.1	72.1	80.4	86.5	88.7	87.0	84.3	73.5	64.3	53.7
	5%	WB	39.0	41.9	50.1	55.4	64.3	71.1	74.2	73.0	68.1	58.2	47.6	38.9
		MCDB	47.8	52.9	63.1	69.8	77.1	84.6	87.3	85.4	81.7	69.9	59.9	48.1
	10%	WB	35.7	38.5	46.7	52.9	61.9	69.3	72.8	71.4	66.0	55.2	44.0	35.4
		MCDB	42.4	46.8	59.1	65.1	73.4	82.3	85.9	83.0	77.9	65.9	54.5	42.4
Mean Daily Temperature Range	5% DB	MDBR	26.2	26.0	28.7	26.8	25.7	25.9	25.8	25.5	28.6	28.6	28.5	26.0
		MCDBR	35.4	37.3	40.5	38.3	33.2	30.9	31.2	30.8	33.9	38.6	39.3	35.2
		MCWBR	23.3	22.9	22.4	19.7	16.0	13.1	12.0	12.2	14.2	19.7	23.0	23.3
	5% WB	MCDBR	33.6	35.8	36.2	33.2	29.0	27.1	26.9	26.6	29.0	32.0	36.9	34.0
		MCWBR	22.6	22.4	20.6	17.9	14.7	13.1	12.3	12.5	13.3	17.7	22.2	22.8
Clear-Sky Solar Irradiance	taub	0.261	0.269	0.294	0.322	0.352	0.366	0.387	0.379	0.346	0.309	0.272	0.262	
	taud	2.543	2.521	2.476	2.417	2.411	2.403	2.388	2.393	2.447	2.519	2.547	2.522	
	Ebn at Noon	292	304	305	301	293	288	280	280	283	284	284	281	
	Edh at Noon	22	26	31	36	36	37	37	36	32	26	22	21	
All-Sky Solar Radiation	RadAvg	725	1008	1367	1690	1967	2255	2231	1938	1571	1114	799	626	
	RadStd	59	66	107	96	135	139	84	89	80	109	50	41	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page